

Sid Mathur's Thematic ETF Screener – Daily Report

2026-08-12

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Action Center

Run date: 2026-08-12 **Universe:** 659 ETFs / 26 clusters

Full PC1: LONG since 2026-08-04 (D+5, 0.2%) **Clusters long:** 10/26 **ETFs long:** 320/659 (49%)

Today's events (cluster level)

$\text{spread_buy} = 5\text{dma} - 50\text{dma}$; $\text{spread_sell} = 5\text{dma} - 20\text{dma}$ (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

BUY (1)

cluster	theme	spread_buy	p_h5	nss
cluster_19	Broad Commodities	1.902	0.94	0.095

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_1	U.S. Mid Cap Equity	3.168	1.000	1.00	1.000	1.00
cluster_11	Healthcare	5.967	1.000	1.00	1.000	0.99
cluster_14	Financials	5.992	1.000	1.00	1.000	0.98
cluster_9	U.S. Dividend Equity	3.507	1.000	1.00	0.876	0.56
cluster_8	Emerging Markets Equity	-0.668	0.206	0.43	0.206	0.42

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-08-04 (5 days, 0.2%)

5dma: 145.469 20dma: 142.906 50dma: 142.441

Spread vs 50dma: +3.028 Spread vs 20dma: +2.563

P(buy crossover within 5 days): 100.0% P(sell): 0.0%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VTV	cluster_6	U.S. Large Cap Value	1.000	83	10.6%	14.48%	0.98
VEA	cluster_3	Developed ex-US Equity	1.000	8	1.8%	8.17%	0.99
BBAX	cluster_7	Asia-Pacific Equity	1.000	19	1.7%	5.76%	0.97
GUNR	cluster_12	Materials	1.000	4	2.2%	5.42%	0.95
KWEB	cluster_13	China Equity	1.000	6	1.0%	3.87%	0.92
IGF	cluster_7	Asia-Pacific Equity	1.000	19	1.7%	3.17%	0.97
IAU	cluster_17	Gold	1.000	2	1.2%	2.38%	0.95
SLV	cluster_17	Gold	1.000	2	1.2%	1.30%	0.95
EWZ	cluster_7	Asia-Pacific Equity	1.000	19	1.7%	1.03%	0.97
URA	cluster_12	Materials	1.000	4	2.2%	0.62%	0.95
GLD	cluster_17	Gold	1.000	2	1.2%	0.06%	0.95
VOO	cluster_2	U.S. Large Cap Equity	0.991	4	0.2%	8.45%	0.99
EWJ	cluster_22	Japan Equity	0.987	3	1.2%	5.67%	0.97
QQQ	cluster_5	U.S. Large Cap Growth	0.801	2	-0.5%	4.84%	0.98
PDBC	cluster_19	Broad Commodities	0.095	0	0.0%	5.35%	0.98

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1	0.983	1.095	0.96
VHT	cluster_11	Healthcare	1	0.995	1.088	0.93
IJH	cluster_1	U.S. Mid Cap Equity	1	1.000	1.000	0.99
SCHD	cluster_9	U.S. Dividend Equity	1	0.555	0.907	0.95
BWX	cluster_25	International Fixed Income	1	0.997	0.751	0.93
INDA	cluster_20	India Equity	1	1.000	0.660	0.91
NOBL	cluster_20	India Equity	1	1.000	0.080	0.91
PPLT	cluster_20	India Equity	1	1.000	0.029	0.91

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
1	cluster_1	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	88	FLAT				1.000	-0.000
3	cluster_3	Developed ex-US Equity	Developed ex-US E...	61	LONG		8	1.8%	1.000	-0.000
6	cluster_6	U.S. Large Cap Value	U.S. Large Cap Va...	49	LONG		83	10.6%	1.000	-0.000
7	cluster_7	Asia-Pacific Equity	Asia-Pacific Equi...	25	LONG		19	1.7%	1.000	-0.000
12	cluster_12	Materials	Materials	17	LONG		4	2.2%	1.000	+0.000
13	cluster_13	China Equity	China Equity	15	LONG		6	1.0%	1.000	-0.000
17	cluster_17	Gold	Gold	12	LONG		2	1.2%	1.000	+0.000
20	cluster_20	India Equity	India Equity	10	FLAT				1.000	-0.000
25	cluster_25	International Fixed Income	International Fix...	6	FLAT				0.997	-0.002
11	cluster_11	Healthcare	Healthcare	17	FLAT				0.995	+0.051
2	cluster_2	U.S. Large Cap Equity	U.S. Large Cap Eq...	78	LONG		4	0.2%	0.991	-0.001
22	cluster_22	Japan Equity	Japan Equity	9	LONG		3	1.2%	0.987	+0.023
14	cluster_14	Financials	Financials	14	FLAT				0.983	-0.012
5	cluster_5	U.S. Large Cap Growth	U.S. Large Cap Gr...	52	LONG		2	-0.5%	0.801	-0.001
9	cluster_9	U.S. Dividend Equity	U.S. Dividend Equ...	23	FLAT				0.555	-0.163
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	FLAT				0.424	+0.004
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG	→LONG	0	0.0%	0.095	+0.241
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.025	+0.061
10	cluster_10	High Yield Bond	High Yield Bond	21	FLAT				-0.200	-0.044
16	cluster_16	Municipal Bond	Municipal Bond	13	FLAT				-0.521	+0.261
15	cluster_15	Real Estate	Real Estate	14	FLAT				-0.614	-0.340
4	cluster_4	Aggregate Fixed Income	Aggregate Fixed I...	58	FLAT				-0.766	-0.107
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				-0.776	+0.122
18	cluster_18	Consumer Staples	Consumer Staples	11	FLAT				-0.883	-0.088
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				-0.917	+0.018
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.999	+0.000

Cluster Details

Cluster 1 – U.S. Mid Cap Equity

Members: 88 **Var explained:** 0.867 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +40.5%)
- Latest sell_cross: 2026-07-22
- Current 252d return: +25.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-22 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Industrials	15	0.1583
U.S. Small Cap Equity	10	0.1195
U.S. Mid Cap Equity	6	0.0721
U.S. Small Cap Value	6	0.0712
U.S. Small Cap Growth	6	0.0700
Consumer Discretionary	6	0.0627
U.S. Mid Cap Value	3	0.0359
U.S. Mid Cap Growth	3	0.0352

Utilities	3	0.0316
Dividend Equity	2	0.0228
Leveraged / Inverse	2	0.0228
U.S. Small Cap Low Volatility	2	0.0217
U.S. Small Cap Fundamental Equity	1	0.0121
Unknown	1	0.0121
U.S. Small Cap ESG Equity	1	0.0121
U.S. Mid/Small Cap Equity	1	0.0121
Mid Cap Value Equities	1	0.0120
U.S. Small/Mid Cap Fundamental Equity	1	0.0120
U.S. Small Cap Multifactor Equity	1	0.0120
U.S. Small/Mid Cap Equity	1	0.0118
Small Cap Value Equities	1	0.0118
U.S. ESG Equity	1	0.0117
U.S. Mid Cap Dividend Equity	1	0.0117
U.S. Extended Market Equity	1	0.0117
U.S. Mid Cap Quality Equity	1	0.0116
U.S. Equity	1	0.0116
Growth Equity	1	0.0116
U.S. Small Cap Dividend Equity	1	0.0115
Value Equity	1	0.0115
U.S. Small Cap Industrials	1	0.0114
U.S. Small Cap Momentum Equity	1	0.0113
U.S. Micro Cap Equity	1	0.0111
U.S. Dividend Equity	1	0.0108
U.S. Small Cap Consumer Discretionary	1	0.0108
U.S. Mid Cap Low Volatility	1	0.0104
Real Estate	1	0.0093

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
IJH	1	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
16	FNX	0.0121	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
19	NUSC	0.0121	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
23	MDY	0.0121	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
31	IVOO	0.0121	Vanguard S&P Mid-Cap 400 ETF		FLAT

35	IJR	0.0121	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
38	IJH	0.0121	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
39	SPMD	0.0121	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
40	SPSM	0.0121	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
41	VB	0.0121	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
60	FNDA	0.0121	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
72	VBR	0.0121	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
11	JHMM	0.0120	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
15	SCHM	0.0120	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
20	PRFZ	0.0120	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
21	SMLF	0.0120	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
24	IJT	0.0120	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
29	SLYG	0.0120	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
32	VIOO	0.0120	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
42	SCHA	0.0120	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
55	EZM	0.0120	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
65	IJJ	0.0120	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
67	IVOV	0.0120	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
68	MDYV	0.0120	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
5	IWM	0.0119	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
17	FYX	0.0119	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
26	VIOG	0.0119	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
33	VTWO	0.0119	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
63	IWN	0.0119	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
71	VTWV	0.0119	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
80	RWK	0.0119	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
54	EES	0.0118	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
56	PSC	0.0118	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
66	IJS	0.0118	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
69	SLYV	0.0118	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
70	VIOV	0.0118	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
87	SDVY	0.0118	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
88	ISMD	0.0118	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
6	FXR	0.0117	First Trust Industrials AlphaDEX	Industrials	FLAT
25	IJK	0.0117	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
27	IVOG	0.0117	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
28	MDYG	0.0117	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
37	VXF	0.0117	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
53	DON	0.0117	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
73	ETHO	0.0117	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
74	RWJ	0.0117	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
36	ISCG	0.0116	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
77	XMHQ	0.0116	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
81	OMFS	0.0116	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
52	DES	0.0115	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT

79	RFV	0.0115	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
86	FYT	0.0115	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
10	IYJ	0.0114	iShares U.S. Industrials ETF	Industrials	FLAT
13	VIS	0.0114	Vanguard Industrials ETF	Industrials	LONG
22	IWO	0.0114	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
30	VTWG	0.0114	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
75	PSCI	0.0114	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
14	FIDU	0.0113	Fidelity MSCI Industrials Index ETF	Industrials	LONG
34	VBK	0.0113	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
43	FXD	0.0113	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
76	PAVE	0.0113	Global X U.S. Infrastructure Developmen...	Industrials	LONG
82	RZV	0.0113	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
85	XSMO	0.0113	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG
18	IWC	0.0111	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
50	RSPD	0.0111	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
1	XLI	0.0110	State Street Industrial Select Sector S...	Industrials	FLAT
9	FIW	0.0110	First Trust Water ETF	Utilities	FLAT
12	REGL	0.0110	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
45	FTXR	0.0110	First Trust Nasdaq Transportation ETF	Industrials	LONG
64	SMMV	0.0110	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
51	SMDV	0.0108	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
84	PSCD	0.0108	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
7	PHO	0.0107	Invesco Water Resources ETF	Utilities	FLAT
59	XTN	0.0107	State Street SPDR S&P Transportation ETF	Industrials	FLAT
78	PKB	0.0107	Invesco Building & Construction ETF	Industrials	FLAT
62	XSLV	0.0106	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
2	XRT	0.0104	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
49	IYT	0.0104	iShares U.S. Transportation ETF	Industrials	LONG
61	XMLV	0.0104	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
44	PEJ	0.0103	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
58	XHB	0.0103	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
8	CGW	0.0100	Invesco S&P Global Water Index ETF	Utilities	FLAT
83	PRN	0.0099	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
46	PPA	0.0095	Invesco Aerospace & Defense ETF	Industrials	LONG
57	XAR	0.0095	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
3	JETS	0.0094	U.S. Global Jets ETF	Industrials	FLAT
48	ITB	0.0094	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG
4	REM	0.0093	iShares Mortgage Real Estate ETF	Real Estate	FLAT
47	ITA	0.0089	iShares U.S. Aerospace & Defense ETF	Industrials	LONG

Cluster 2 – U.S. Large Cap Equity

Members: 78 **Var explained:** 0.873 **Signal:** LONG since 2026-08-05 (4 days, 0.2%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +19.7%)
- Latest sell_cross: 2026-07-20
- Current 252d return: +19.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-07-20)

**Category breakdown**

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1226
U.S. Total Market Equity	7	0.0957
Global Equity	7	0.0938
Technology	5	0.0618
Consumer Discretionary	5	0.0607
Asset Allocation	4	0.0498
U.S. Equity	3	0.0399
U.S. Mega Cap Equity	3	0.0399
U.S. Multifactor Equity	3	0.0398
U.S. Quality / Factor Equity	3	0.0394
U.S. Mid Cap Growth	3	0.0392

U.S. Momentum Equity	3	0.0352
Communication Services	3	0.0348
U.S. ESG Equity	2	0.0272
Momentum Equity	2	0.0244
Convertible Bond	2	0.0233
Commodity	1	0.0136
Large Cap Blend Equities	1	0.0135
Large Cap Value Equities	1	0.0135
Dividend Equity	1	0.0132
U.S. Equal Weight Equity	1	0.0130
U.S. Dividend Equity	1	0.0129
Growth Equity	1	0.0127
U.S. Moat Equity	1	0.0123
IPO Equity	1	0.0122
U.S. Mid Cap Momentum Equity	1	0.0120
Alternatives	1	0.0119
U.S. Small Cap Momentum Equity	1	0.0115
International IPO Equity	1	0.0104
Israel Equity	1	0.0100

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
VOO	1.007	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
48	IWV	0.0137	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
49	IYY	0.0137	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
55	IWB	0.0137	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
60	VTHR	0.0137	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
63	VONE	0.0137	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
67	ITOT	0.0137	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
69	SCHB	0.0137	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
70	SCHX	0.0137	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
73	VTI	0.0137	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
5	SPY	0.0136	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
12	URTH	0.0136	iShares MSCI World ETF	Global Equity	LONG
44	JHML	0.0136	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
47	SUSA	0.0136	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG

52	SPYX	0.0136	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
54	ESGU	0.0136	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
61	GSLC	0.0136	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
65	VV	0.0136	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
66	ILCB	0.0136	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
68	IVV	0.0136	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
71	SPTM	0.0136	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
72	VOO	0.0136	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
7	ACWI	0.0135	iShares MSCI ACWI ETF	Global Equity	LONG
13	CRBN	0.0135	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
15	VT	0.0135	Vanguard Total World Stock Index ETF	Global Equity	LONG
38	TILT	0.0135	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
46	DSI	0.0135	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
50	LRGF	0.0135	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
56	IWL	0.0135	iShares Russell Top 200 ETF	U.S. Equity	LONG
62	EPS	0.0135	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
64	MGC	0.0135	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
8	AOA	0.0134	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
14	SPGM	0.0134	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
57	QUAL	0.0134	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
51	OEF	0.0133	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
36	IMCG	0.0132	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
42	QDF	0.0132	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
43	CATH	0.0132	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	VOT	0.0131	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
53	XLG	0.0131	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
58	QUS	0.0131	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
11	AOR	0.0130	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
18	QQEW	0.0130	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
19	XT	0.0130	iShares Future Exponential Technologies...	Technology	LONG
28	IWP	0.0130	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
41	IOO	0.0130	iShares Global 100 ETF	Global Equity	LONG
59	SPHQ	0.0130	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	FLAT
40	SPHB	0.0129	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
45	DGRW	0.0129	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
21	FTC	0.0128	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
26	RPG	0.0127	Invesco S&P 500 Pure Growth ETF	Growth Equity	FLAT
78	OMFL	0.0127	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
24	IYC	0.0126	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
4	ROBO	0.0125	ROBO Global Robotics and Automation Ind...	Technology	LONG
22	PDP	0.0124	Invesco Dorsey Wright Momentum ETF	Momentum Equity	FLAT
31	VCR	0.0124	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
34	FDIS	0.0124	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
3	BOTZ	0.0123	Global X Robotics & Artificial Intellig...	Technology	LONG
37	MOAT	0.0123	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT

75	VFMO	0.0123	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	FLAT
1	XLY	0.0122	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	AOM	0.0122	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
23	FPX	0.0122	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
16	FV	0.0120	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
27	XMMO	0.0120	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	FLAT
6	QAI	0.0119	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
25	CWB	0.0119	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
30	MTUM	0.0118	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	FLAT
32	VOX	0.0117	Vanguard Communication Services ETF	Communication Services	LONG
33	FCOM	0.0117	Fidelity MSCI Communication Services In...	Communication Services	LONG
39	DWAS	0.0115	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
29	ICVT	0.0114	iShares Convertible Bond ETF	Convertible Bond	FLAT
2	XLC	0.0113	State Street Communication Services Sel...	Communication Services	LONG
17	PRNT	0.0113	3D Printing (The) ETF	Technology	LONG
9	AOK	0.0112	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
74	QMOM	0.0111	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
77	RTH	0.0111	VanEck Retail ETF	Consumer Discretionary	LONG
20	FPXI	0.0104	First Trust International Equity Opport...	International IPO Equity	FLAT
76	EIS	0.0100	iShares MSCI Israel ETF	Israel Equity	FLAT

Cluster 3 – Developed ex-US Equity

Members: 61 **Var explained:** 0.894 **Signal:** LONG since 2026-07-30 (8 days, 1.8%)

Signal logic:

- Latest buy_cross: 2026-07-30 – filter **PASS** (252d at date = +21.9%)
- Latest sell_cross: 2026-06-25
- Current 252d return: +22.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-30 > last sell at 2026-06-25)



Category breakdown

category	n_members	abs_weight
International Equity	19	0.3157
Developed ex-US Equity	9	0.1529
Global ex-US Equity	5	0.0848
Europe Equity	4	0.0675
International Dividend Equity	4	0.0659
Alternatives	4	0.0616
Eurozone Equity	3	0.0490
Commodity	1	0.0172
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162
Italy Equity	1	0.0159
Europe Financials	1	0.0158
United Kingdom Equity	1	0.0157
Netherlands Equity	1	0.0156
Sweden Equity	1	0.0156
Spain Equity	1	0.0151
Austria Equity	1	0.0148
Belgium Equity	1	0.0145
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.021

ticker	rep_wt	name	category	etf_signal
VEA	0.958	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0173	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
51	IEFA	0.0173	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
40	GSIE	0.0172	Goldman Sachs ActiveBeta International ...	Commodity	LONG
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	LONG
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
25	TLTD	0.0171	FlexShares Morningstar Developed Market...	International Equity	LONG
32	INTF	0.0171	iShares International Equity Factor ETF	International Equity	LONG
6	IEV	0.0170	iShares Europe ETF	Europe Equity	LONG
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	LONG
19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	LONG
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
49	VGK	0.0170	Vanguard FTSEEurope ETF	Europe Equity	LONG
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	LONG
14	EZU	0.0168	iShares MSCI Eurozone ETF	Eurozone Equity	LONG
23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
38	VYMI	0.0168	Vanguard International High Dividend Yi...	International Dividend E...	LONG
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	LONG
22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	LONG
26	JPIN	0.0167	JPMorgan Diversified Return Internation...	International Equity	LONG
34	IVLU	0.0167	iShares MSCI Intl Value Factor ETF	International Equity	LONG
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	LONG
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	LONG
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	LONG

36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	LONG
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	LONG
31	IMTM	0.0163	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	LONG
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	LONG
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	LONG
21	GWX	0.0161	State Street SPDR S&P International Sma...	International Equity	LONG
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	LONG
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	LONG
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	FLAT
15	IDV	0.0158	iShares International Select Dividend E...	International Dividend E...	LONG
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	LONG
13	EWU	0.0157	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
27	DBEF	0.0157	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	LONG
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
42	HDEF	0.0156	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	LONG
58	EWN	0.0156	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	LONG
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	FLAT
11	EWSP	0.0151	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWAO	0.0148	iShares MSCI Austria ETF	Austria Equity	LONG
60	EWBK	0.0145	iShares MSCI Belgium ETF	Belgium Equity	LONG
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	LONG

Cluster 4 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.831 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +1.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3284
Investment Grade Corporate Bond	13	0.2303
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0306
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
27	USIG	0.0181	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
10	CORP	0.0178	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
33	BSCQ	0.0155	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0151	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 5 – U.S. Large Cap Growth

Members: 52 **Var explained:** 0.838 **Signal:** LONG since 2026-08-07 (2 days, -0.5%)

Signal logic:

- Latest buy_cross: 2026-08-07 – filter **PASS** (252d at date = +28.5%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +27.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-07 > last sell at 2026-07-17)



Category breakdown

category	n_members	abs_weight
Technology	29	0.5521
U.S. Large Cap Growth	11	0.2257
Growth Equity	2	0.0403
Financials	2	0.0347
U.S. Equity	1	0.0207
U.S. Mega Cap Growth	1	0.0204
Options Income	1	0.0185
U.S. Small Cap Technology	1	0.0183
IPO Equity	1	0.0183
Trend-Following Equity	1	0.0180
Consumer Discretionary	1	0.0172
U.S. Trend-Following Equity	1	0.0158

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.023

ticker	rep_wt	name	category	etf_signal
QQQ	1.059	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
30	IGM	0.0208	iShares Expanded Tech Sector ETF	Technology	LONG
12	QQQ	0.0207	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
34	ONEQ	0.0207	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
43	ILCG	0.0207	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
32	IYW	0.0206	iShares U.S. Technology ETF	Technology	LONG
38	VGT	0.0206	Vanguard Information Tech ETF	Technology	LONG
40	FTEC	0.0206	Fidelity MSCI Information Technology In...	Technology	LONG
24	XNTK	0.0205	State Street SPDR NYSE Technology ETF	Technology	FLAT
33	NULG	0.0205	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
36	IWF	0.0205	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
41	VONG	0.0205	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
44	IUSG	0.0205	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
47	VUG	0.0205	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
14	QTEC	0.0204	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
37	IYW	0.0204	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
39	VOOG	0.0204	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
42	MGK	0.0204	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
45	SCHG	0.0204	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
46	SPYG	0.0204	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
1	XLK	0.0203	State Street Technology Select Sector S...	Technology	LONG
19	FXL	0.0203	First Trust Technology AlphaDEX	Technology	LONG
35	IWY	0.0203	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
6	AIQ	0.0202	Global X Artificial Intelligence & Tech...	Technology	LONG
31	IXN	0.0202	iShares Global Tech ETF	Technology	LONG
29	PWB	0.0199	Invesco Large Cap Growth ETF	Growth Equity	LONG
16	TDIV	0.0193	First Trust NASDAQ Technology Dividend ...	Technology	LONG
52	QTUM	0.0191	Defiance Quantum ETF	Technology	LONG
3	SMH	0.0190	VanEck Semiconductor ETF	Technology	FLAT
23	FDN	0.0190	First Trust DJ Internet Index Fund	Technology	LONG
2	SOXX	0.0187	iShares PHLX SOX Semiconductor Sector I...	Technology	FLAT
5	SKYY	0.0186	First Trust Cloud Computing ETF	Technology	LONG
11	ARKQ	0.0186	ARK Autonomous Technology & Robotics ETF	Technology	LONG
17	XSD	0.0185	State Street SPDR S&P Semiconductor ETF	Technology	FLAT
22	PNQI	0.0185	Invesco Nasdaq Internet ETF	Technology	FLAT
28	QYLD	0.0185	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
13	NXTG	0.0184	First Trust Indxx NextG ETF	Technology	FLAT
21	IPO	0.0183	Renaissance IPO ETF	IPO Equity	FLAT
49	PSCT	0.0183	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	FLAT
51	PTF	0.0183	Invesco Dorsey Wright Technology Moment...	Technology	LONG
4	IGV	0.0182	iShares Expanded Tech-Software Sector E...	Technology	FLAT
15	PSI	0.0182	Invesco Semiconductors ETF	Technology	FLAT
50	FTXL	0.0182	First Trust Nasdaq Semiconductor ETF	Technology	FLAT
10	ARKW	0.0181	ARK Next Generation Internet ETF	Technology	FLAT

26	FINX	0.0180	Global X FinTech ETF	Financials	FLAT
27	PTNQ	0.0180	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
9	ARKK	0.0178	ARK Innovation ETF	Technology	LONG
20	CIBR	0.0175	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
8	HACK	0.0172	Amplify Cybersecurity ETF	Technology	FLAT
18	IBUY	0.0172	Amplify Online Retail ETF	Consumer Discretionary	FLAT
25	IPAY	0.0168	Amplify Digital Payments ETF	Financials	FLAT
7	BLOK	0.0164	Amplify Blockchain Technology ETF	Technology	FLAT
48	PTLC	0.0158	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG

Cluster 6 – U.S. Large Cap Value

Members: 49 **Var explained:** 0.920 **Signal:** LONG since 2026-04-13 (83 days, 10.6%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +24.2%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +23.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Value	10	0.2095
U.S. Dividend Equity	8	0.1651
U.S. Multifactor Equity	3	0.0616
U.S. Mid Cap Equity	3	0.0613
U.S. Value Factor Equity	3	0.0601
U.S. Fundamental Equity	2	0.0420
U.S. Equal Weight Equity	2	0.0416
U.S. Mid Cap Value	2	0.0414
U.S. Equity	2	0.0411
Value Equity	2	0.0405
U.S. Low Volatility Equity	2	0.0385
U.S. Revenue Weighted Equity	1	0.0207
Large Cap Blend Equities	1	0.0207
U.S. Mega Cap Value	1	0.0206
U.S. Size Factor Equity	1	0.0206
U.S. Large Cap Equity	1	0.0203
U.S. Buyback Equity	1	0.0202
Dividend Equity	1	0.0197
U.S. Quality / Factor Equity	1	0.0194
Global Equity	1	0.0183
Communication Services	1	0.0167

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.001

ticker	rep_wt	name	category	etf_signal
VTV	1.021	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
18	IWD	0.0211	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
25	VONV	0.0211	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
2	RSP	0.0210	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
7	PRF	0.0210	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
10	CFA	0.0210	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
14	FNDX	0.0210	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
28	ILCV	0.0210	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
31	SCHV	0.0210	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG

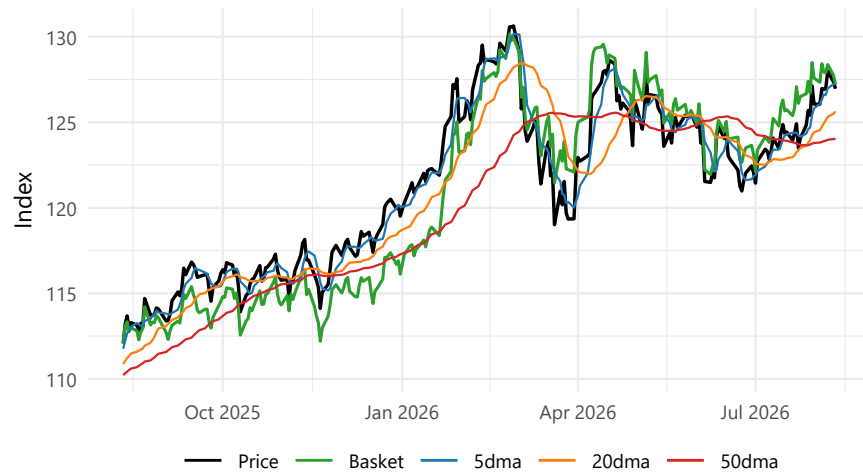
20	IVE	0.0209	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
21	JPUS	0.0209	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
24	VOOV	0.0209	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
30	IUSV	0.0209	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
32	SPYV	0.0209	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
34	VTV	0.0209	Vanguard Value ETF	U.S. Large Cap Value	LONG
41	DTD	0.0209	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
45	DGRO	0.0209	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
12	NULV	0.0208	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
16	IWX	0.0208	iShares Russell Top 200 Value ETF	Value Equity	LONG
40	DLN	0.0208	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
4	FEX	0.0207	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
8	RWL	0.0207	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
27	VYM	0.0207	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
35	IWS	0.0207	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
36	VOE	0.0207	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
38	PFM	0.0207	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
22	SIZE	0.0206	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	FLAT
26	MGV	0.0206	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	FLAT
46	VIG	0.0206	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
48	JPME	0.0206	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
9	AIVL	0.0205	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
15	EQAL	0.0205	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
29	IMCB	0.0205	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
13	FDVV	0.0204	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
17	ONEV	0.0204	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
19	IWR	0.0204	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
33	VO	0.0204	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
49	JVAL	0.0204	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
42	DIA	0.0203	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	FLAT
3	PKW	0.0202	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
6	RDVY	0.0200	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
5	PWV	0.0198	Invesco Large Cap Value ETF	Value Equity	LONG
11	CFO	0.0197	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
39	OUSL	0.0197	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
37	FTCS	0.0194	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
44	LGLV	0.0193	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
23	VLUE	0.0192	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
43	USMV	0.0192	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
1	ACWV	0.0183	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
47	IYZ	0.0167	iShares U.S. Telecommunications ETF	Communication Services	LONG

Cluster 7 – Asia-Pacific Equity

Members: 25 **Var explained:** 0.670 **Signal:** LONG since 2026-07-15 (19 days, 1.7%)

Signal logic:

- Latest buy_cross: 2026-07-15 – filter **PASS** (252d at date = +13.2%)
- Latest sell_cross: 2026-06-23
- Current 252d return: +13.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-15 > last sell at 2026-06-23)

**Category breakdown**

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0916
International Equity	2	0.0888
International Dividend Equity	2	0.0886
Real Estate	2	0.0853
Global Equity	2	0.0848
Emerging Markets Fixed Income	2	0.0776
Currency	2	0.0749
Australia Equity	1	0.0453
Canada Equity	1	0.0430
Switzerland Equity	1	0.0400
Singapore Equity	1	0.0387

South Africa Equity	1	0.0383
New Zealand Equity	1	0.0366
Latin America Equity	1	0.0365
Mexico Equity	1	0.0347
Consumer Discretionary	1	0.0337
Chile Equity	1	0.0309
Brazil Equity	1	0.0308

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
EWZ	0.087	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.267	iShares Global Infrastructure ETF	Global Equity	FLAT
BBAX	0.486	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
9	EPP	0.0458	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	LONG
25	BBAX	0.0458	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
5	EWA	0.0453	iShares MSCI Australia Index Fund	Australia Equity	LONG
13	IDLV	0.0453	Invesco S&P International Developed Low...	International Equity	LONG
12	DWX	0.0447	State Street SPDR S&P International Div...	International Dividend E...	LONG
4	PID	0.0440	Invesco International Dividend Achiever...	International Dividend E...	LONG
14	EFAV	0.0435	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
15	VNQI	0.0432	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
8	EWC	0.0430	iShares MSCI Canada Index Fund	Canada Equity	LONG
10	NFRA	0.0430	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
3	RWX	0.0421	State Street SPDR Dow Jones Internation...	Real Estate	LONG
11	IGF	0.0418	iShares Global Infrastructure ETF	Global Equity	FLAT
6	EWL	0.0400	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
21	FXA	0.0399	Invesco CurrencyShares Australian Dolla...	Currency	LONG
16	EBND	0.0389	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
7	EWS	0.0387	iShares MSCI Singapore ETF	Singapore Equity	LONG
17	EMLC	0.0387	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
23	EZA	0.0383	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
22	ENZL	0.0366	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
20	ILF	0.0365	iShares Latin America 40 ETF	Latin America Equity	LONG
24	FXC	0.0350	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
18	EWX	0.0347	iShares MSCI Mexico ETF	Mexico Equity	LONG
1	ESPO	0.0337	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT

19	ECH	0.0309	iShares MSCI Chile ETF	Chile Equity	LONG
2	EWZ	0.0308	iShares MSCI Brazil ETF	Brazil Equity	FLAT

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.870 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +28.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Emerging Markets Equity	19	0.7729
Asia Equity	2	0.0815
Asia ex-Japan Equity	1	0.0419
Taiwan Equity	1	0.0362

South Korea Equity	1	0.0340
Global Dividend Equity	1	0.0335

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.034

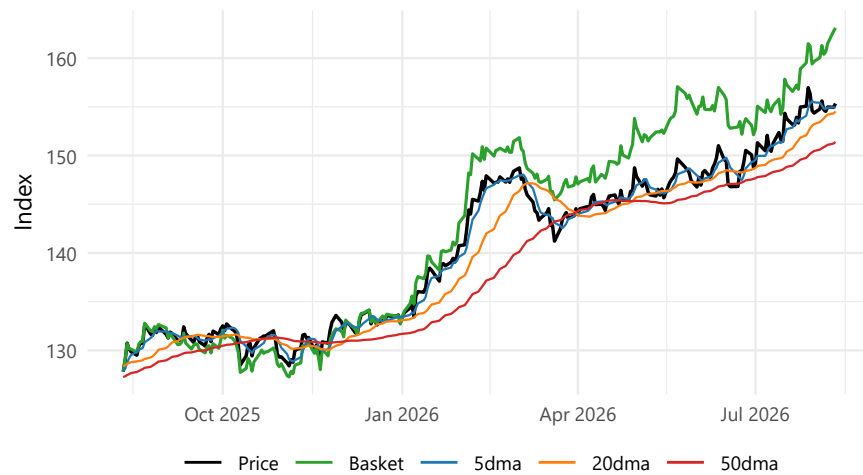
ticker	rep_wt	name	category	etf_signal
IEMG	0.918	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0426	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
20	ESGE	0.0423	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
5	AAXJ	0.0419	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
11	EEMA	0.0415	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0412	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
17	FNDE	0.0409	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
12	PXH	0.0408	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
10	AIA	0.0403	iShares Asia 50 ETF	Asia Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	FLAT
4	FEM	0.0389	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0363	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	LONG
9	EWT	0.0362	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0349	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0340	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0335	Global X SuperDividend ETF	Global Dividend Equity	FLAT

Cluster 9 – U.S. Dividend Equity**Members:** 23 **Var explained:** 0.825 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-22 – filter **PASS** (252d at date = +22.4%)
- Latest sell_cross: 2026-06-23
- Current 252d return: +21.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-04-22)

**Category breakdown**

category	n_members	abs_weight
U.S. Dividend Equity	13	0.5935
MLP / Energy Infrastructure	4	0.1403
Value Equity	2	0.0902
U.S. Low Volatility Equity	2	0.0848
Dividend Equity	1	0.0460
U.S. Defensive Equity	1	0.0453

Replication basket – SCHD (cor=0.95) basket lead at lag 1: 0.016

ticker	rep_wt	name	category	etf_signal
SCHD	0.907	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
17	DVY	0.0471	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
22	SPYD	0.0467	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
7	DHS	0.0465	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
16	SDOG	0.0465	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
9	SDY	0.0460	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
20	SPHD	0.0460	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
21	ONEY	0.0460	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
13	FTA	0.0459	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
6	FDL	0.0456	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
12	SCHD	0.0456	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
8	CDC	0.0453	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
5	FVD	0.0452	First Trust VL Dividend	U.S. Dividend Equity	LONG
15	DIV	0.0449	Global X Super Dividend ETF	U.S. Dividend Equity	FLAT
18	RDIV	0.0448	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
11	HDV	0.0444	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
14	PEY	0.0443	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
19	RPV	0.0442	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
10	LVHD	0.0440	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
23	SPLV	0.0408	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT
1	EMLP	0.0407	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	FLAT
4	MLPX	0.0349	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	FLAT
2	AML	0.0328	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
3	MLPA	0.0319	Global X MLP ETF	MLP / Energy Infrastruct...	LONG

Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.764 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-07-07

- Current 252d return: +3.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5158
Preferred Stock	7	0.3057
Emerging Markets Fixed Income	4	0.1786

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.046

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0527	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT

19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT
14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0517	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0512	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0472	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0458	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
8	VRP	0.0369	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.756 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +36.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
VHT	1.088	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0656	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0655	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0648	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0642	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0640	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0625	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0623	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	FLAT
6	FBT	0.0597	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	FLAT
7	XHE	0.0572	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0565	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0544	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0536	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0531	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0506	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0473	iShares U.S. Health Care Providers ETF	Healthcare	FLAT

Cluster 12 – Materials

Members: 17 **Var explained:** 0.750 **Signal:** LONG since 2026-08-05 (4 days, 2.2%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +35.2%)
- Latest sell_cross: 2026-06-22
- Current 252d return: +35.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-06-22)



Category breakdown

category	n_members	abs_weight
Materials	13	0.7920
Consumer Staples	1	0.0584
Energy	1	0.0566
Norway Equity	1	0.0512
Industrial Metals	1	0.0419

Replication basket – URA+GUNR (cor=0.95) basket lead at lag 1: 0.037

ticker	rep_wt	name	category	etf_signal
URA	0.102	Global X Uranium ETF	Materials	LONG
GUNR	0.889	FlexShares Global Upstream Natural Re...	Materials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	GNR	0.0654	State Street SPDR S&P Global Natural Re...	Materials	LONG
6	MXI	0.0649	iShares Global Materials ETF	Materials	LONG
7	IYM	0.0644	iShares U.S. Basic Materials ETF	Materials	LONG
10	GUNR	0.0643	FlexShares Global Upstream Natural Reso...	Materials	LONG
12	PICK	0.0628	iShares MSCI Global Select Metals & Min...	Materials	LONG
9	VAW	0.0623	Vanguard Materials ETF	Materials	LONG
17	NANR	0.0615	State Street SPDR S&P North American Na...	Materials	LONG
1	XLB	0.0611	State Street Materials Select Sector SP...	Materials	LONG
8	RSPM	0.0610	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
13	SLX	0.0608	VanEck Steel ETF	Materials	LONG
4	XME	0.0605	State Street SPDR S&P Metals & Mining E...	Materials	LONG
3	COPX	0.0587	Global X Copper Miners ETF	Materials	LONG
5	MOO	0.0584	VanEck Agribusiness ETF	Consumer Staples	FLAT
16	IGE	0.0566	iShares North American Natural Resource...	Energy	LONG
14	NORW	0.0512	Global X MSCI Norway ETF	Norway Equity	LONG
2	URA	0.0442	Global X Uranium ETF	Materials	LONG
15	DBB	0.0419	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 13 – China Equity

Members: 15 **Var explained:** 0.795 **Signal:** LONG since 2026-08-03 (6 days, 1.0%)

Signal logic:

- Latest buy_cross: 2026-08-03 – filter **PASS** (252d at date = +2.6%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +2.4% – filter **ALLOWES** new entries today
- Decision: **LONG** (effective buy at 2026-08-03 > last sell at 2026-05-20)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8765
Emerging Markets Equity	1	0.0644
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.010

ticker	rep_wt	name	category	etf_signal
KWEB	0.562	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0740	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0738	State Street SPDR S&P China ETF	China Equity	LONG
1	MCHI	0.0732	iShares MSCI China ETF	China Equity	LONG
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0660	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0653	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0644	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0597	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
14	CNXT	0.0588	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0582	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 14 – Financials

Members: 14 **Var explained:** 0.872 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +15.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9382
Global Equity	1	0.0618

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
XLF	1.095	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0757	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0756	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0754	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	LONG
8	IYG	0.0749	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0742	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0734	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0713	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0696	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0630	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0618	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +13.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-06 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	FLAT
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0722	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	FLAT
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0716	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	FLAT
4	REZ	0.0696	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	FLAT
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.820 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-07-10
- Current 252d return: +4.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-10 more recent than last effective buy 2026-05-28)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0794	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0794	iShares New York Muni Bond ETF	Municipal Bond	FLAT

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0770	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0721	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	FLAT
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Gold

Members: 12 **Var explained:** 0.863 **Signal:** LONG since 2026-08-07 (2 days, 1.2%)

Signal logic:

- Latest buy_cross: 2026-08-07 – filter **PASS** (252d at date = +36.2%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +35.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-07 > last sell at 2026-05-20)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1723
Silver	2	0.1617

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
GLD	0.019	SPDR Gold Shares	Gold	LONG
IAU	0.755	iShares Gold Trust Shares	Gold	LONG
SLV	0.412	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0865	Invesco DB Precious Metals Fund	Precious Metals	LONG
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	LONG
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	LONG
2	GDXJ	0.0840	VanEck Junior Gold Miners ETF	Materials	LONG
3	GLD	0.0835	SPDR Gold Shares	Gold	LONG
9	IAU	0.0834	iShares Gold Trust Shares	Gold	LONG
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	LONG
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	LONG
6	SIL	0.0828	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0808	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-26 – filter **PASS** (252d at date = +9.9%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +3.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2026-06-26)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5742
Utilities	4	0.3507
U.S. Small Cap Consumer Staples	1	0.0752

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.016

ticker	rep_wt	name	category	etf_signal
XLU	0.363	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0984	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0984	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0976	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0951	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG

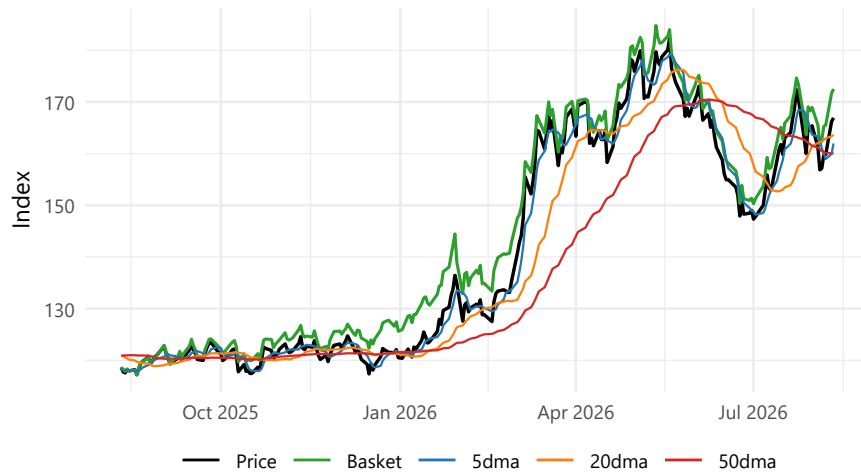
3	KXI	0.0947	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0900	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0887	iShares U.S. Utilities ETF	Utilities	FLAT
9	FUTY	0.0875	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0874	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0871	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0752	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.770 **Signal:** LONG since 2026-08-11 (0 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-08-11 – filter **PASS** (252d at date = +41.2%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +41.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-11 > last sell at 2026-08-06)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6174
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0494
Natural Gas	1	0.0332

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.079

ticker	rep_wt	name	category	etf_signal
PDBC	1.125	Invesco Optimum Yield Diversified Com...	Broad Commodities	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1057	Invesco DB Commodity Index Tracking Fund	Broad Commodities	FLAT
6	GSG	0.1057	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1056	Invesco Optimum Yield Diversified Commo...	Broad Commodities	FLAT
9	COMT	0.1049	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.0996	United States Oil Fund	Oil	LONG
5	DBO	0.0984	Invesco DB Oil Fund	Oil	LONG
7	DJP	0.0984	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
4	FTGC	0.0972	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
3	DBA	0.0494	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
11	UNG	0.0332	United States Natural Gas Fund LP	Natural Gas	FLAT

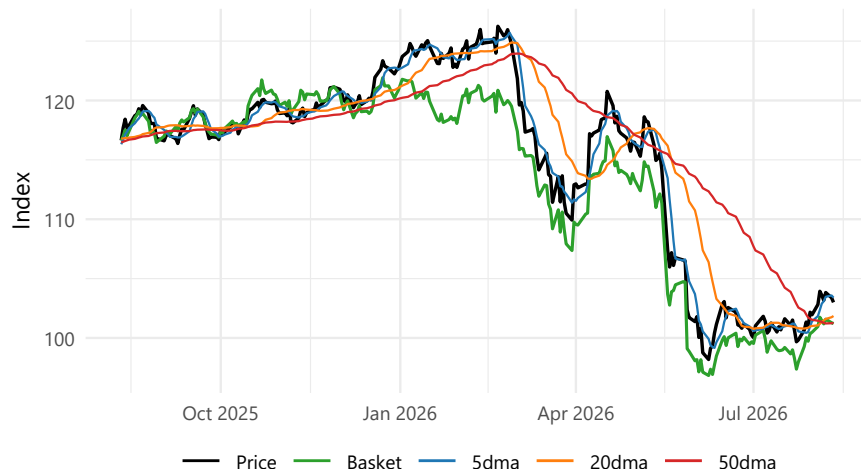
Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-03 – filter **BLOCKED** (252d at date = -10.3%; needs > 0)
- Latest effective buy: 2026-05-06 – delayed re-arm from blocked cross (252d at fire = +4.7%)
- Latest sell_cross: 2026-07-27
- Current 252d return: -11.6% – filter **BLOCKS** new entries today

- Decision: **FLAT** (last sell 2026-07-27 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4679
Malaysia Equity	1	0.1126
Indonesia Equity	1	0.1038
Philippines Equity	1	0.1014
Vietnam Equity	1	0.0750
Turkey Equity	1	0.0607
U.S. Dividend Equity	1	0.0548
Platinum	1	0.0239

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1573	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1557	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1549	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1126	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	EIDO	0.1038	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1014	iShares MSCI Philippines ETF	Philippines Equity	LONG
10	VNM	0.0750	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0607	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0548	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0239	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.904 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: -0.9% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.268	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.760	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1036	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1035	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1024	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

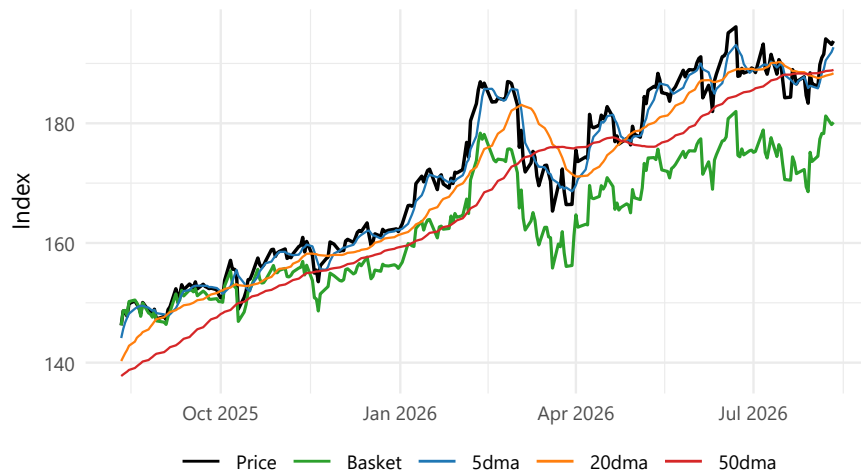
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0989	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0884	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** LONG since 2026-08-06 (3 days, 1.2%)

Signal logic:

- Latest buy_cross: 2026-08-06 – filter **PASS** (252d at date = +35.6%)
- Latest sell_cross: 2026-07-28
- Current 252d return: +32.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-06 > last sell at 2026-07-28)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7805

Asia-Pacific Equity	2	0.2195
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.045

ticker	rep_wt	name	category	etf_signal
EWJ	0.907	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
6	DBJP	0.1118	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1110	Franklin FTSE Japan Hedged ETF	Japan Equity	LONG
2	DXJ	0.1095	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1081	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1079	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.555 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +3.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3622
Interest Rate Hedged Fixed Income	3	0.3243
Floating Rate Fixed Income	3	0.3136

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.217	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.298	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.123	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1251	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG

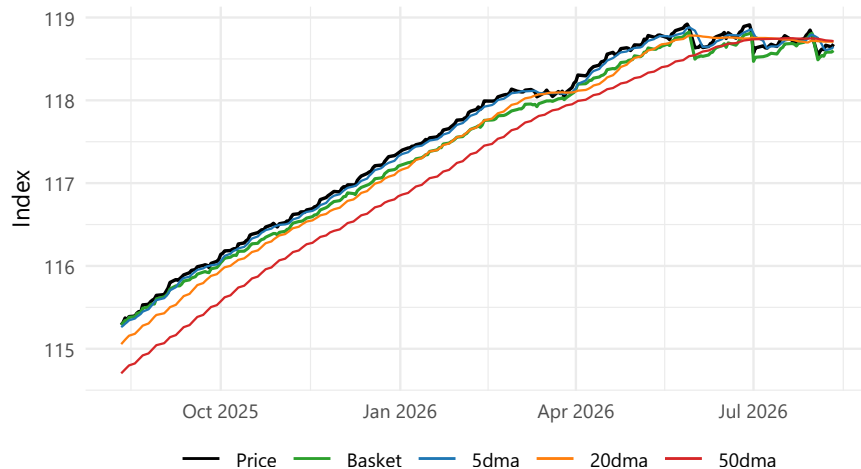
3	BKLN	0.1208	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1163	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1142	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1085	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1055	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1046	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1039	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1012	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.461 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-07-29 – filter **PASS** (252d at date = +3.3%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +3.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2026-07-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.86) basket lead at lag 1: 0.048

ticker	rep_wt	name	category	etf_signal
SHV	0.253	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.172	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
GBIL	0.125	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
ICSH	0.400	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1324	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1309	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1155	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1124	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
4	NEAR	0.1101	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.1065	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1057	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0948	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.0917	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-04 – filter **BLOCKED** (252d at date = -1.4%; needs > 0)
- Latest effective buy: 2026-05-29 – delayed re-arm from blocked cross (252d at fire = +0.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: -2.0% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-29)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409
International Fixed Income	2	0.3591

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.024

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	LONG
5	FXF	0.1651	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1447	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.626 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-08-05
- Current 252d return: -33.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
SLX	cluster_12	Materials	2026-07-30	8	2.7%	0.00	1.00
GNR	cluster_12	Materials	2026-07-27	11	5.5%	0.00	1.00
PICK	cluster_12	Materials	2026-08-05	4	0.7%	0.00	1.00
RSPM	cluster_12	Materials	2026-07-27	11	3.6%	0.00	1.00
DBB	cluster_12	Materials	2026-08-04	5	1.6%	0.00	1.00
MXI	cluster_12	Materials	2026-08-05	4	1.0%	0.00	1.00
COPX	cluster_12	Materials	2026-08-05	4	2.2%	0.00	1.00
NANR	cluster_12	Materials	2026-07-24	12	6.0%	0.00	1.00
XLB	cluster_12	Materials	2026-07-28	10	1.7%	0.00	1.00
VAW	cluster_12	Materials	2026-08-06	3	2.2%	0.00	1.00
IYM	cluster_12	Materials	2026-08-06	3	3.1%	0.00	1.00
NORW	cluster_12	Materials	2026-08-05	4	2.3%	0.00	1.00
GUNR	cluster_12	Materials	2026-07-27	11	3.7%	0.00	1.00
IGE	cluster_12	Materials	2026-07-23	13	4.5%	0.00	1.00
XME	cluster_12	Materials	2026-08-10	1	0.2%	0.00	1.00
URA	cluster_12	Materials	2026-08-11	0	0.0%	0.00	0.97
TLTD	cluster_3	Developed ex-US Equity	2026-07-30	8	2.2%	0.00	1.00
IQLT	cluster_3	Developed ex-US Equity	2026-07-01	28	5.4%	0.00	1.00
JPIN	cluster_3	Developed ex-US Equity	2026-07-27	11	3.0%	0.00	1.00
FNDC	cluster_3	Developed ex-US Equity	2026-08-04	5	1.0%	0.00	1.00
EWQ	cluster_3	Developed ex-US Equity	2026-07-30	8	2.6%	0.00	1.00
EPOL	cluster_3	Developed ex-US Equity	2026-07-08	24	11.6%	0.00	1.00
DWM	cluster_3	Developed ex-US Equity	2026-07-30	8	1.5%	0.00	1.00
IPKW	cluster_3	Developed ex-US Equity	2026-07-10	22	5.7%	0.00	1.00
GSIE	cluster_3	Developed ex-US Equity	2026-07-02	27	4.5%	0.00	1.00
QEFA	cluster_3	Developed ex-US Equity	2026-07-06	26	3.5%	0.00	1.00
DLS	cluster_3	Developed ex-US Equity	2026-08-03	6	1.7%	0.00	1.00
FEZ	cluster_3	Developed ex-US Equity	2026-07-30	8	3.1%	0.00	1.00
FNDF	cluster_3	Developed ex-US Equity	2026-07-31	7	2.0%	0.00	1.00
INTF	cluster_3	Developed ex-US Equity	2026-07-27	11	3.6%	0.00	1.00
VGK	cluster_3	Developed ex-US Equity	2026-07-01	28	5.2%	0.00	1.00
RODM	cluster_3	Developed ex-US Equity	2026-07-27	11	1.8%	0.00	1.00
PXF	cluster_3	Developed ex-US Equity	2026-08-03	6	1.9%	0.00	1.00
EWD	cluster_3	Developed ex-US Equity	2026-07-30	8	1.1%	0.00	1.00
SCHC	cluster_3	Developed ex-US Equity	2026-08-05	4	1.3%	0.00	1.00
EFA	cluster_3	Developed ex-US Equity	2026-07-30	8	1.7%	0.00	1.00
ESGD	cluster_3	Developed ex-US Equity	2026-07-30	8	1.7%	0.00	1.00

FEP	cluster_3	Developed	ex-US	Equity	2026-07-31	7	1.1%	0.00	1.00
SCZ	cluster_3	Developed	ex-US	Equity	2026-08-04	5	0.9%	0.00	1.00
IEFA	cluster_3	Developed	ex-US	Equity	2026-07-30	8	1.8%	0.00	1.00
IEV	cluster_3	Developed	ex-US	Equity	2026-07-27	11	3.8%	0.00	1.00
IEUR	cluster_3	Developed	ex-US	Equity	2026-07-30	8	1.3%	0.00	1.00
EWG	cluster_3	Developed	ex-US	Equity	2026-07-31	7	2.7%	0.00	1.00
VYMI	cluster_3	Developed	ex-US	Equity	2026-07-08	24	4.8%	0.00	1.00
IVLU	cluster_3	Developed	ex-US	Equity	2026-07-27	11	3.0%	0.00	1.00
EZU	cluster_3	Developed	ex-US	Equity	2026-07-31	7	2.8%	0.00	1.00
VIGI	cluster_3	Developed	ex-US	Equity	2026-06-30	29	6.1%	0.00	1.00
EFV	cluster_3	Developed	ex-US	Equity	2026-07-08	24	5.1%	0.00	1.00
HEDJ	cluster_3	Developed	ex-US	Equity	2026-08-03	6	0.8%	0.00	1.00
HEZU	cluster_3	Developed	ex-US	Equity	2026-07-30	8	2.7%	0.00	1.00
EUFN	cluster_3	Developed	ex-US	Equity	2026-06-12	40	8.5%	0.00	1.00
GWX	cluster_3	Developed	ex-US	Equity	2026-08-06	3	1.7%	0.00	1.00
EWK	cluster_3	Developed	ex-US	Equity	2026-07-29	9	-0.8%	0.00	1.00
IQDF	cluster_3	Developed	ex-US	Equity	2026-08-04	5	0.1%	0.00	1.00
IDV	cluster_3	Developed	ex-US	Equity	2026-07-28	10	2.0%	0.00	1.00
EWU	cluster_3	Developed	ex-US	Equity	2026-07-16	18	2.8%	0.00	1.00
SPDW	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.5%	0.00	1.00
VEA	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.5%	0.00	1.00
EW0	cluster_3	Developed	ex-US	Equity	2026-07-31	7	2.9%	0.00	1.00
EFG	cluster_3	Developed	ex-US	Equity	2026-08-05	4	1.2%	0.00	1.00
SCHF	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.4%	0.00	1.00
VXUS	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.5%	0.00	1.00
IXUS	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.4%	0.00	1.00
VSS	cluster_3	Developed	ex-US	Equity	2026-08-07	2	0.1%	0.00	1.00
CWI	cluster_3	Developed	ex-US	Equity	2026-08-04	5	0.3%	0.00	1.00
ACWX	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.3%	0.00	1.00
VEU	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.4%	0.00	1.00
HDEF	cluster_3	Developed	ex-US	Equity	2026-07-09	23	4.0%	0.01	0.99
IMTM	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.9%	0.00	0.98
HFXI	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.3%	0.00	0.98
VIDI	cluster_3	Developed	ex-US	Equity	2026-08-10	1	-0.1%	0.00	0.96
EWN	cluster_3	Developed	ex-US	Equity	2026-08-05	4	0.9%	0.00	0.89
IWD	cluster_6	U.S.	Large	Cap Value	2026-04-10	84	16.3%	0.00	1.00
VONV	cluster_6	U.S.	Large	Cap Value	2026-04-10	84	16.2%	0.00	1.00
IWX	cluster_6	U.S.	Large	Cap Value	2026-04-10	84	17.5%	0.00	1.00
FNDX	cluster_6	U.S.	Large	Cap Value	2026-04-09	85	12.5%	0.00	1.00
VIG	cluster_6	U.S.	Large	Cap Value	2026-04-14	82	9.7%	0.00	1.00
DLN	cluster_6	U.S.	Large	Cap Value	2026-04-13	83	9.7%	0.00	1.00
RWL	cluster_6	U.S.	Large	Cap Value	2026-04-10	84	13.8%	0.00	1.00
PKW	cluster_6	U.S.	Large	Cap Value	2026-05-20	56	10.3%	0.00	1.00
DTD	cluster_6	U.S.	Large	Cap Value	2026-04-13	83	9.4%	0.00	1.00
PRF	cluster_6	U.S.	Large	Cap Value	2026-04-09	85	13.2%	0.00	1.00

DGRO	cluster_6	U.S. Large Cap Value	2026-04-13	83	10.2%	0.00	1.00
VTV	cluster_6	U.S. Large Cap Value	2026-04-13	83	11.2%	0.00	1.00
ACWV	cluster_6	U.S. Large Cap Value	2026-07-06	26	3.4%	0.00	1.00
RSP	cluster_6	U.S. Large Cap Value	2026-04-14	82	10.5%	0.00	1.00
AIVL	cluster_6	U.S. Large Cap Value	2026-04-10	84	11.6%	0.00	1.00
OUSA	cluster_6	U.S. Large Cap Value	2026-07-01	28	5.3%	0.00	1.00
JVAL	cluster_6	U.S. Large Cap Value	2026-07-30	8	3.6%	0.00	1.00
FEX	cluster_6	U.S. Large Cap Value	2026-04-09	85	10.7%	0.00	1.00
IYZ	cluster_6	U.S. Large Cap Value	2026-08-06	3	0.6%	0.00	1.00
FTCS	cluster_6	U.S. Large Cap Value	2026-06-26	31	7.0%	0.00	1.00
SCHV	cluster_6	U.S. Large Cap Value	2026-08-04	5	0.0%	0.00	1.00
PWV	cluster_6	U.S. Large Cap Value	2026-04-02	89	15.8%	0.04	0.96
LGLV	cluster_6	U.S. Large Cap Value	2026-06-26	31	3.0%	0.09	0.91
VLUE	cluster_6	U.S. Large Cap Value	2026-08-07	2	0.1%	0.00	0.65
GOAU	cluster_17	Gold	2026-08-06	3	8.0%	0.00	1.00
GDX	cluster_17	Gold	2026-08-06	3	7.4%	0.00	1.00
GDXJ	cluster_17	Gold	2026-08-06	3	8.4%	0.00	1.00
SGOL	cluster_17	Gold	2026-08-07	2	0.7%	0.00	1.00
IAU	cluster_17	Gold	2026-08-07	2	0.6%	0.00	1.00
GLD	cluster_17	Gold	2026-08-07	2	0.6%	0.00	1.00
SIL	cluster_17	Gold	2026-08-07	2	2.6%	0.00	1.00
DBP	cluster_17	Gold	2026-08-07	2	0.8%	0.00	1.00
GLTR	cluster_17	Gold	2026-08-10	1	-0.8%	0.00	1.00
SILJ	cluster_17	Gold	2026-08-07	2	3.2%	0.00	1.00
SLV	cluster_17	Gold	2026-08-10	1	-1.4%	0.00	1.00
SIVR	cluster_17	Gold	2026-08-10	1	-1.4%	0.00	1.00
SDVY	cluster_1	U.S. Mid Cap Equity	2026-06-08	44	7.6%	0.00	1.00
PPA	cluster_1	U.S. Mid Cap Equity	2026-07-27	11	3.9%	0.00	1.00
ITA	cluster_1	U.S. Mid Cap Equity	2026-07-24	12	4.9%	0.00	1.00
RSPD	cluster_1	U.S. Mid Cap Equity	2026-07-28	10	1.4%	0.00	1.00
SMLF	cluster_1	U.S. Mid Cap Equity	2026-08-03	6	2.3%	0.00	1.00
ISCG	cluster_1	U.S. Mid Cap Equity	2026-08-04	5	0.3%	0.00	1.00
XAR	cluster_1	U.S. Mid Cap Equity	2026-08-06	3	3.3%	0.00	1.00
VXF	cluster_1	U.S. Mid Cap Equity	2026-08-05	4	1.4%	0.00	1.00
VTWO	cluster_1	U.S. Mid Cap Equity	2026-08-04	5	-0.2%	0.00	1.00
IWM	cluster_1	U.S. Mid Cap Equity	2026-08-04	5	-0.2%	0.00	1.00
PAVE	cluster_1	U.S. Mid Cap Equity	2026-08-05	4	0.1%	0.00	1.00
IVOG	cluster_1	U.S. Mid Cap Equity	2026-08-06	3	1.8%	0.00	1.00
MDYG	cluster_1	U.S. Mid Cap Equity	2026-08-06	3	1.9%	0.00	1.00
IJK	cluster_1	U.S. Mid Cap Equity	2026-08-06	3	1.9%	0.00	1.00
VIS	cluster_1	U.S. Mid Cap Equity	2026-08-05	4	-0.3%	0.00	0.99
FIDU	cluster_1	U.S. Mid Cap Equity	2026-08-05	4	-0.2%	0.00	0.99
ISMD	cluster_1	U.S. Mid Cap Equity	2026-04-09	85	18.7%	0.01	0.99
REGL	cluster_1	U.S. Mid Cap Equity	2026-06-12	40	4.9%	0.01	0.99
IJS	cluster_1	U.S. Mid Cap Equity	2026-04-10	84	12.8%	0.01	0.99

IWC	cluster_1	U.S. Mid Cap Equity	2026-08-05	4	0.7%	0.00	0.99
ITB	cluster_1	U.S. Mid Cap Equity	2026-07-28	10	0.2%	0.00	0.95
VBK	cluster_1	U.S. Mid Cap Equity	2026-08-07	2	0.3%	0.00	0.95
IWO	cluster_1	U.S. Mid Cap Equity	2026-08-06	3	1.3%	0.00	0.93
VTWG	cluster_1	U.S. Mid Cap Equity	2026-08-06	3	1.2%	0.00	0.92
SCHM	cluster_1	U.S. Mid Cap Equity	2026-08-07	2	-0.3%	0.00	0.84
XHB	cluster_1	U.S. Mid Cap Equity	2026-08-07	2	-0.5%	0.01	0.80
XRT	cluster_1	U.S. Mid Cap Equity	2026-05-28	51	4.7%	0.21	0.79
SMMV	cluster_1	U.S. Mid Cap Equity	2026-06-25	32	4.5%	0.25	0.75
SCHA	cluster_1	U.S. Mid Cap Equity	2026-08-07	2	-0.2%	0.01	0.57
XSMO	cluster_1	U.S. Mid Cap Equity	2026-08-07	2	0.1%	0.03	0.38
FTXR	cluster_1	U.S. Mid Cap Equity	2026-08-05	4	-1.3%	0.50	-0.28
IYT	cluster_1	U.S. Mid Cap Equity	2026-08-06	3	0.8%	0.90	-0.65
EWA	cluster_7	Asia-Pacific Equity	2026-07-20	16	5.0%	0.00	1.00
EPP	cluster_7	Asia-Pacific Equity	2026-07-15	19	3.8%	0.00	1.00
EZA	cluster_7	Asia-Pacific Equity	2026-08-05	4	2.6%	0.00	1.00
BBAX	cluster_7	Asia-Pacific Equity	2026-07-14	20	4.3%	0.00	1.00
EWG	cluster_7	Asia-Pacific Equity	2026-07-10	22	4.9%	0.00	1.00
EBND	cluster_7	Asia-Pacific Equity	2026-07-31	7	0.6%	0.00	1.00
EWS	cluster_7	Asia-Pacific Equity	2026-06-17	37	14.6%	0.00	1.00
ECH	cluster_7	Asia-Pacific Equity	2026-08-05	4	-1.4%	0.00	1.00
EFAV	cluster_7	Asia-Pacific Equity	2026-07-16	18	3.6%	0.00	1.00
EWL	cluster_7	Asia-Pacific Equity	2026-07-28	10	1.3%	0.00	1.00
EMLC	cluster_7	Asia-Pacific Equity	2026-08-05	4	0.0%	0.00	1.00
FXA	cluster_7	Asia-Pacific Equity	2026-08-04	5	0.2%	0.01	0.99
EWV	cluster_7	Asia-Pacific Equity	2026-07-31	7	-0.7%	0.01	0.99
ENZL	cluster_7	Asia-Pacific Equity	2026-07-07	25	4.4%	0.01	0.99
IDLV	cluster_7	Asia-Pacific Equity	2026-07-10	22	2.8%	0.03	0.97
DWX	cluster_7	Asia-Pacific Equity	2026-07-09	23	2.6%	0.10	0.90
RWX	cluster_7	Asia-Pacific Equity	2026-07-16	18	1.2%	0.19	0.81
VNQI	cluster_7	Asia-Pacific Equity	2026-07-20	16	0.0%	0.40	0.60
NFRA	cluster_7	Asia-Pacific Equity	2026-07-16	18	0.2%	0.56	0.43
PID	cluster_7	Asia-Pacific Equity	2026-07-16	18	-0.5%	0.78	0.18
ILF	cluster_7	Asia-Pacific Equity	2026-07-22	14	-4.7%	0.83	0.13
EPHE	cluster_20	India Equity	2026-08-03	6	-1.0%	0.08	0.92
EWM	cluster_20	India Equity	2026-07-30	8	-0.7%	0.28	0.72
GXC	cluster_13	China Equity	2026-07-31	7	0.3%	0.00	1.00
MCHI	cluster_13	China Equity	2026-08-04	5	-0.8%	0.00	1.00
CXSE	cluster_13	China Equity	2026-08-07	2	-1.4%	0.00	0.99
KURE	cluster_13	China Equity	2026-07-06	26	9.2%	0.04	0.96
EWI	cluster_13	China Equity	2026-07-23	13	0.2%	0.51	0.49
FXE	cluster_25	International Fixed Inc...	2026-08-04	5	0.1%	0.00	1.00
FXB	cluster_25	International Fixed Inc...	2026-07-31	7	0.1%	0.16	0.84
IYH	cluster_11	Healthcare	2026-05-20	56	14.5%	0.02	0.98
VHT	cluster_11	Healthcare	2026-05-19	57	15.2%	0.02	0.98

XLV	cluster_11	Healthcare	2026-05-21	55	13.4%	0.02	0.98
FHLC	cluster_11	Healthcare	2026-05-19	57	15.2%	0.02	0.98
PPH	cluster_11	Healthcare	2026-05-21	55	7.9%	0.22	0.78
PTH	cluster_11	Healthcare	2026-06-17	37	13.6%	0.25	0.75
RTH	cluster_2	U.S. Large Cap Equity	2026-07-29	9	3.9%	0.00	1.00
DGRW	cluster_2	U.S. Large Cap Equity	2026-08-03	6	2.1%	0.00	1.00
PRNT	cluster_2	U.S. Large Cap Equity	2026-08-05	4	1.8%	0.00	1.00
JHML	cluster_2	U.S. Large Cap Equity	2026-08-03	6	1.7%	0.00	1.00
IOO	cluster_2	U.S. Large Cap Equity	2026-08-03	6	1.2%	0.00	1.00
GSLC	cluster_2	U.S. Large Cap Equity	2026-08-03	6	1.8%	0.00	1.00
URTH	cluster_2	U.S. Large Cap Equity	2026-08-03	6	1.9%	0.00	1.00
LRGF	cluster_2	U.S. Large Cap Equity	2026-08-03	6	1.9%	0.00	1.00
SPGM	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.3%	0.00	1.00
EPS	cluster_2	U.S. Large Cap Equity	2026-08-03	6	1.5%	0.00	1.00
ACWI	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.2%	0.00	1.00
VT	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.2%	0.00	1.00
SCHX	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.0%	0.00	1.00
SPTM	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
VTI	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.0%	0.00	1.00
AOA	cluster_2	U.S. Large Cap Equity	2026-08-05	4	0.1%	0.00	1.00
CATH	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
VTHR	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
SCHB	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.0%	0.00	1.00
VONE	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.0%	0.00	1.00
IWV	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.0%	0.00	1.00
IYY	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
VOO	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
IWB	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.0%	0.00	1.00
ITOT	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.0%	0.00	1.00
SPYX	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.2%	0.00	1.00
SPY	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
ILCB	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.0%	0.00	1.00
IVV	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
VV	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
SUSA	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.0%	0.00	1.00
CRBN	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
ESGU	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.0%	0.00	1.00
DSI	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.4%	0.00	1.00
OEF	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.2%	0.00	1.00
MGC	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
IWL	cluster_2	U.S. Large Cap Equity	2026-08-04	5	-0.1%	0.00	1.00
AOR	cluster_2	U.S. Large Cap Equity	2026-08-05	4	0.0%	0.00	1.00
IMCG	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.7%	0.00	1.00
IYC	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.7%	0.00	1.00
QQEW	cluster_2	U.S. Large Cap Equity	2026-08-04	5	0.7%	0.00	0.99

AOM	cluster_2	U.S. Large Cap Equity	2026-08-06	3	0.1%	0.00	0.99
XLG	cluster_2	U.S. Large Cap Equity	2026-08-05	4	-0.3%	0.00	0.99
XLY	cluster_2	U.S. Large Cap Equity	2026-08-05	4	0.5%	0.01	0.98
FDIS	cluster_2	U.S. Large Cap Equity	2026-08-05	4	0.0%	0.01	0.98
VOT	cluster_2	U.S. Large Cap Equity	2026-08-05	4	0.3%	0.00	0.97
VCR	cluster_2	U.S. Large Cap Equity	2026-08-05	4	-0.0%	0.01	0.97
BOTZ	cluster_2	U.S. Large Cap Equity	2026-08-07	2	-0.3%	0.00	0.97
AOK	cluster_2	U.S. Large Cap Equity	2026-08-06	3	0.1%	0.00	0.96
QAI	cluster_2	U.S. Large Cap Equity	2026-08-06	3	0.5%	0.00	0.95
ROBO	cluster_2	U.S. Large Cap Equity	2026-08-07	2	-0.5%	0.00	0.94
IWP	cluster_2	U.S. Large Cap Equity	2026-08-07	2	0.5%	0.00	0.91
XT	cluster_2	U.S. Large Cap Equity	2026-08-07	2	0.2%	0.00	0.91
XLC	cluster_2	U.S. Large Cap Equity	2026-08-06	3	0.1%	0.20	0.72
FTC	cluster_2	U.S. Large Cap Equity	2026-08-10	1	0.5%	0.00	0.54
FV	cluster_2	U.S. Large Cap Equity	2026-08-10	1	0.2%	0.00	0.51
SPHB	cluster_2	U.S. Large Cap Equity	2026-08-10	1	0.5%	0.01	0.43
DWAS	cluster_2	U.S. Large Cap Equity	2026-08-10	1	0.1%	0.06	0.29
VOX	cluster_2	U.S. Large Cap Equity	2026-08-07	2	-0.5%	0.44	0.12
FCOM	cluster_2	U.S. Large Cap Equity	2026-08-07	2	-0.5%	0.44	0.12
IPAC	cluster_22	Japan Equity	2026-08-03	6	3.1%	0.00	1.00
EWJV	cluster_22	Japan Equity	2026-07-06	26	2.9%	0.00	1.00
EWJ	cluster_22	Japan Equity	2026-08-05	4	1.2%	0.00	1.00
BBJP	cluster_22	Japan Equity	2026-08-05	4	1.2%	0.00	1.00
DXJ	cluster_22	Japan Equity	2026-08-05	4	1.9%	0.01	0.98
HEWJ	cluster_22	Japan Equity	2026-08-07	2	0.6%	0.01	0.86
DBJP	cluster_22	Japan Equity	2026-08-10	1	0.2%	0.01	0.85
FLJH	cluster_22	Japan Equity	2026-08-10	1	0.5%	0.01	0.79
VPL	cluster_22	Japan Equity	2026-08-10	1	0.7%	0.00	0.64
XLF	cluster_14	Financials	2026-04-13	83	11.9%	0.00	1.00
FNCL	cluster_14	Financials	2026-04-13	83	11.8%	0.01	0.99
KCE	cluster_14	Financials	2026-07-07	25	5.4%	0.01	0.99
FXO	cluster_14	Financials	2026-06-05	45	10.7%	0.02	0.98
IAT	cluster_14	Financials	2026-05-20	56	13.5%	0.27	0.73
KIE	cluster_14	Financials	2026-06-10	42	9.5%	0.75	0.25
SKYY	cluster_5	U.S. Large Cap Growth	2026-07-30	8	15.0%	0.00	1.00
PTLC	cluster_5	U.S. Large Cap Growth	2026-08-04	5	-0.2%	0.00	1.00
FDN	cluster_5	U.S. Large Cap Growth	2026-08-03	6	2.6%	0.00	1.00
SCHG	cluster_5	U.S. Large Cap Growth	2026-08-04	5	0.4%	0.00	1.00
IUSG	cluster_5	U.S. Large Cap Growth	2026-08-05	4	-0.2%	0.00	1.00
SPYG	cluster_5	U.S. Large Cap Growth	2026-08-05	4	-0.2%	0.00	0.99
IVW	cluster_5	U.S. Large Cap Growth	2026-08-05	4	-0.3%	0.00	0.99
VOOG	cluster_5	U.S. Large Cap Growth	2026-08-05	4	-0.2%	0.00	0.99
VUG	cluster_5	U.S. Large Cap Growth	2026-08-05	4	-0.2%	0.00	0.99
MGK	cluster_5	U.S. Large Cap Growth	2026-08-05	4	-0.1%	0.01	0.98
FXL	cluster_5	U.S. Large Cap Growth	2026-08-06	3	2.6%	0.00	0.97

VGT	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.5%	0.00	0.97
FTEC	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.5%	0.00	0.96
ONEQ	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.3%	0.01	0.96
NULG	cluster_5	U.S. Large Cap Growth	2026-08-05	4	-0.4%	0.00	0.96
TDIV	cluster_5	U.S. Large Cap Growth	2026-08-07	2	-0.3%	0.00	0.94
ILCG	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.1%	0.01	0.93
IYW	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.2%	0.00	0.92
VONG	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.1%	0.01	0.91
IWF	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.1%	0.01	0.91
IGM	cluster_5	U.S. Large Cap Growth	2026-08-06	3	0.3%	0.00	0.89
IWY	cluster_5	U.S. Large Cap Growth	2026-08-07	2	-0.9%	0.02	0.87
ARKQ	cluster_5	U.S. Large Cap Growth	2026-08-11	0	0.0%	0.00	0.83
XLK	cluster_5	U.S. Large Cap Growth	2026-08-07	2	-1.0%	0.00	0.81
QQQ	cluster_5	U.S. Large Cap Growth	2026-08-07	2	-0.6%	0.01	0.68
ARKK	cluster_5	U.S. Large Cap Growth	2026-08-10	1	0.2%	0.02	0.68
PTNQ	cluster_5	U.S. Large Cap Growth	2026-08-07	2	-0.7%	0.01	0.68
IXN	cluster_5	U.S. Large Cap Growth	2026-08-07	2	-0.7%	0.00	0.67
QYLD	cluster_5	U.S. Large Cap Growth	2026-08-07	2	0.1%	0.04	0.63
AIQ	cluster_5	U.S. Large Cap Growth	2026-08-10	1	-0.5%	0.00	0.59
QTEC	cluster_5	U.S. Large Cap Growth	2026-08-10	1	-0.1%	0.01	0.43
PWB	cluster_5	U.S. Large Cap Growth	2026-08-07	2	-0.7%	0.03	0.41
QTUM	cluster_5	U.S. Large Cap Growth	2026-04-10	84	31.1%	0.00	0.30
PTF	cluster_5	U.S. Large Cap Growth	2026-04-07	87	13.7%	0.05	-0.03
SCHD	cluster_9	U.S. Dividend Equity	2026-07-07	25	5.3%	0.00	1.00
FTA	cluster_9	U.S. Dividend Equity	2026-05-21	55	9.2%	0.00	1.00
SDY	cluster_9	U.S. Dividend Equity	2026-05-20	56	6.6%	0.01	0.99
FVD	cluster_9	U.S. Dividend Equity	2026-06-10	42	5.9%	0.01	0.99
HDV	cluster_9	U.S. Dividend Equity	2026-06-25	32	5.6%	0.06	0.94
FDL	cluster_9	U.S. Dividend Equity	2026-07-16	18	1.8%	0.11	0.89
AMPL	cluster_9	U.S. Dividend Equity	2026-07-10	22	5.0%	0.73	0.27
SPHD	cluster_9	U.S. Dividend Equity	2026-06-08	44	5.2%	0.90	0.09
MLPA	cluster_9	U.S. Dividend Equity	2026-07-10	22	3.2%	0.95	0.05
PXH	cluster_8	Emerging Markets Equity	2026-07-31	7	0.4%	0.00	1.00
FNDE	cluster_8	Emerging Markets Equity	2026-07-30	8	0.9%	0.00	1.00
DVYE	cluster_8	Emerging Markets Equity	2026-07-24	12	1.0%	0.00	1.00
SCHE	cluster_8	Emerging Markets Equity	2026-08-05	4	0.2%	0.00	1.00
VWO	cluster_8	Emerging Markets Equity	2026-08-06	3	0.3%	0.00	0.98
SPEM	cluster_8	Emerging Markets Equity	2026-08-06	3	0.4%	0.00	0.97
FEM	cluster_8	Emerging Markets Equity	2026-08-07	2	-0.6%	0.01	0.92
GMF	cluster_8	Emerging Markets Equity	2026-08-06	3	0.8%	0.00	0.92
DEM	cluster_8	Emerging Markets Equity	2026-08-07	2	-0.5%	0.00	0.88
EWT	cluster_8	Emerging Markets Equity	2026-08-11	0	0.0%	0.02	0.42
AIA	cluster_8	Emerging Markets Equity	2026-07-10	22	-2.1%	0.06	0.07
USO	cluster_19	Broad Commodities	2026-08-11	0	0.0%	0.89	-0.11
DBO	cluster_19	Broad Commodities	2026-08-11	0	0.0%	0.92	-0.16

BKLN	cluster_23	Bank Loan / Floating Ra...	2026-08-05	4	0.2%	0.01	0.99
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-07-30	8	0.1%	0.05	0.95
FTSL	cluster_23	Bank Loan / Floating Ra...	2026-08-07	2	0.1%	0.09	0.77
PFXF	cluster_10	High Yield Bond	2026-08-10	1	-0.2%	0.00	0.92
BSCQ	cluster_4	Aggregate Fixed Income	2026-08-07	2	0.0%	0.16	0.50
LDUR	cluster_4	Aggregate Fixed Income	2026-07-31	7	-0.2%	0.32	0.21
GSY	cluster_24	Cash / Ultra-Short Fixe...	2026-08-07	2	0.0%	0.10	0.64
KXI	cluster_18	Consumer Staples	2026-07-28	10	-2.0%	0.31	0.67
PSCC	cluster_18	Consumer Staples	2026-06-10	42	7.8%	0.34	0.66
FSTA	cluster_18	Consumer Staples	2026-07-28	10	-2.3%	0.33	0.62
VDC	cluster_18	Consumer Staples	2026-07-28	10	-2.3%	0.34	0.60
XLP	cluster_18	Consumer Staples	2026-07-27	11	-0.8%	0.48	0.42
IYK	cluster_18	Consumer Staples	2026-06-09	43	2.2%	0.82	0.06

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
TLTD	cluster_3	Developed ex-US Equity	1.00	1.00
SLX	cluster_12	Materials	1.00	1.00
IWD	cluster_6	U.S. Large Cap Value	1.00	1.00
XMHQ	cluster_1	U.S. Mid Cap Equity	1.00	1.00
EWA	cluster_7	Asia-Pacific Equity	1.00	1.00
PSP	cluster_14	Financials	1.00	1.00
GOAU	cluster_17	Gold	1.00	1.00
SKYY	cluster_5	U.S. Large Cap Growth	1.00	1.00
QUS	cluster_2	U.S. Large Cap Equity	1.00	1.00
IPAC	cluster_22	Japan Equity	1.00	1.00
GXC	cluster_13	China Equity	1.00	1.00
IHI	cluster_11	Healthcare	1.00	1.00
SCHD	cluster_9	U.S. Dividend Equity	1.00	1.00
PXH	cluster_8	Emerging Markets Equity	1.00	1.00
BKLN	cluster_23	Bank Loan / Floating Rate	1.00	0.99
PFXF	cluster_10	High Yield Bond	0.92	0.92
KXI	cluster_18	Consumer Staples	0.98	0.67
GSY	cluster_24	Cash / Ultra-Short Fixed ...	0.75	0.64
COMT	cluster_19	Broad Commodities	1.00	0.54
BSCQ	cluster_4	Aggregate Fixed Income	0.66	0.50

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
PSK	cluster_10	High Yield Bond	1.00	-1.00
FUTY	cluster_18	Consumer Staples	1.00	-1.00
IGF	cluster_7	Asia-Pacific Equity	1.00	-1.00
SCHP	cluster_21	Inflation-Protected Fixed...	1.00	-1.00
MLPX	cluster_9	U.S. Dividend Equity	1.00	-1.00
UUP	cluster_26	Leveraged / Inverse	1.00	-0.99

FLTR	cluster_23	Bank Loan / Floating Rate	1.00	-0.99
TLT	cluster_4	Aggregate Fixed Income	0.98	-0.98
XTN	cluster_1	U.S. Mid Cap Equity	0.97	-0.97
NURE	cluster_15	Real Estate	1.00	-0.96
UNG	cluster_19	Broad Commodities	0.91	-0.91
VNLA	cluster_24	Cash / Ultra-Short Fixed ...	0.89	-0.86
HYMB	cluster_16	Municipal Bond	0.78	-0.78
BLOK	cluster_5	U.S. Large Cap Growth	0.65	-0.64
FPX	cluster_2	U.S. Large Cap Equity	0.59	-0.59
SDIV	cluster_8	Emerging Markets Equity	0.97	-0.16
TUR	cluster_20	India Equity	0.46	-0.13
CNXT	cluster_13	China Equity	0.08	-0.08
MOO	cluster_12	Materials	1.00	-0.04
IAI	cluster_14	Financials	0.78	0.04

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.481	0.075	0.00017	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.305	0.096	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.647	0.204	0.00036	
FXY	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.729	0.095	0.00023	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.765	0.027	0.00007	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	FLAT	0.596	0.169	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.737	0.119	0.00030	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.737	0.139	0.00038	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.696	0.162	0.00044	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.713	0.177	0.00044	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.698	0.185	0.00045	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	FLAT	0.501	0.218	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_4	58	LONG	0.847	0.124	0.00036	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.885	0.024	0.00001	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.885	0.054	0.00018	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_4	58	FLAT	0.875	0.160	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.821	0.202	0.00050	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_4	58	FLAT	0.831	0.234	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.448	0.272	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.877	0.246	0.00046	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.443	0.327	0.00083	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	LONG	0.788	0.267	0.00072	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	FLAT	0.877	0.262	0.00055	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	LONG	0.745	0.359	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.497	0.349	0.00102	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.866	0.317	0.00071	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	FLAT	0.857	0.255	0.00073	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	LONG	0.556	0.404	0.00107	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.917	0.225	0.00057	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.902	0.238	0.00065	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20